

Session 5 · Deck 5.A6

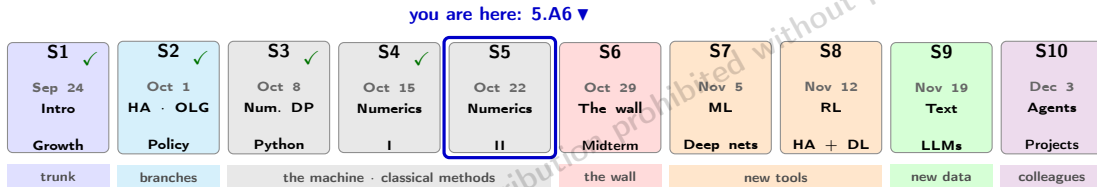
Worked Example: The RBC Model by Perturbation

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Build it on paper → solve it on the machine → push it to the frontier with AI

Where We Are



Build it on paper → solve it on the machine → push it to the frontier with AI

Block A: 5.A1 what perturbation computes · 5.A2 the model and the parameter · 5.A3 first order · 5.A4 linear RE systems · 5.A5 higher order, risk, pruning · **5.A6** the worked example.

The Same Model, a Third Way

3.A4 solved this model with every choice made the crudest way. 4.B5 solved it with the Session 4 toolkit. Both put values on a grid and iterated to a fixed point.

This deck solves it a third way, and nothing about the procedure resembles the first two: no grid, no iteration, no maximization. Five steps, all of them linear algebra.

4.B5	here
grid in k	one point, k^*
Rouwenhorst	ρ and σ directly
Π	
golden section	no maximization at all
Howard sweeps	one QZ decomposition

The model

$$c + k' = zk^\alpha + (1 - \delta)k, \quad \ln z' = \rho \ln z + \sigma \varepsilon'$$

with $\ln$ utility. Quarterly: $\alpha = 0.33$, $\beta = 0.99$, $\delta = 0.025$, $\rho = 0.95$, $\sigma = 0.007$.

$\delta \neq 1$, so there is no closed-form policy. The benchmarks are the steady state, the Euler residual, and 4.B5's own solution.

What “Perturbation” Commits Us To

Six decisions, and they are different decisions from the six of 3.A2:

Choice	Here
expansion point	deterministic steady state
variable	log deviations
order	first, then second
root selection	stable, by Blanchard–Kahn
derivatives	numerical (Dynare would use analytic)
domain	<i>unspecified</i> — and that is the problem

The last row

A grid method tells you its domain: it is the grid. Perturbation does not. Nothing in the output says where the answer stops being usable — which is exactly what we will have to measure.

So the deck’s structure is forced: solve in three slides, then spend four finding out where the solution is valid.

Step 1: The Steady State, and the Linearized System

```
kss = ((1/beta - 1 + delta)/alpha)**(1/(alpha-1))
yss = kss**alpha # 3.0153
css = yss - delta*kss # 2.3066
kappa = beta*alpha*kss**(alpha-1) # 0.03475

# A E_t s' = B s, s = (khat, z, chat)
A = np.array([[kss, 0.0, 0.0],
              [0.0, 1.0, 0.0],
              [-kappa*(alpha-1), -kappa, 1.0]])
B = np.array([[alpha*yss + (1-delta)*kss, yss, -css],
              [0.0, rho, 0.0],
              [0.0, 0.0, 1.0]])
```

Ledger pert_first_order. $k^* = 28.348419$; 4.B5 reports 28.3484.

Check it first

$k^* = 28.3484$ — identical to 4.B5's, to every digit. Two completely different programs agreeing on the one quantity they share is the cheapest test available.

Rows of A and B : resource constraint, shock process, Euler equation. Every entry is an elasticity evaluated at the steady state.

Step 2: One QZ Decomposition

```
def stable(a, b):  
    return np.abs(b) < np.abs(a)    #  $|\omega| < 1$   
  
S, T, aa, bb, Q, Z = ordqz(A, B, sort=stable,  
                           output="real")  
  
omega = np.abs(bb)/np.abs(aa)  
n_unstable = np.sum(omega > 1.0)    # 1  
  
n_x = 2                                #  $khat, z$   
Z11, Z21 = Z[:n_x, :n_x], Z[n_x:, :n_x]  
F = Z21 @ inv(Z11)                    #  $chat = F x$   
P = Z11 @ solve(S[:n_x, :n_x],  
                T[:n_x, :n_x]) @ inv(Z11)
```

Ledger blanchard_kahn. scipy.linalg.ordqz; Klein (2000).

The whole solve

One library call, then two matrix products. $|\omega| = 0.950, 0.962, 1.050$: one root outside the unit circle, one jump variable. Blanchard–Kahn holds.

If the count were wrong, `inv(Z11)` would fail — the algorithm cannot form a solution that does not exist. Do not wrap this in a try.

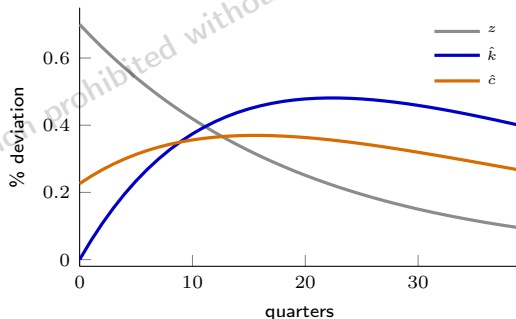
Step 3: Read Off the Policy, and the Impulse Response

The solution is four numbers:

$\hat{k}'$ on $\hat{k}$	0.962061
$\hat{k}'$ on z	0.080097
$\hat{c}$ on $\hat{k}$	0.590408
$\hat{c}$ on z	0.322850

Everything else follows. The impulse response to a one-standard-deviation innovation is a loop over 40 quarters — no simulation, no Monte Carlo.

Ledger fig_irf and pert_first_order.



Reading the Impulse Response

The shock itself decays at $\rho = 0.95$: half of it is gone after 13.5 quarters.

Consumption jumps immediately, by 0.226%, and keeps rising to a peak of **0.370%** — it does not track the shock, because capital is still accumulating.

Capital starts at zero, by construction: it is predetermined. It peaks at **0.481%** in quarter **22** — five and a half years after a shock that was half gone in three.

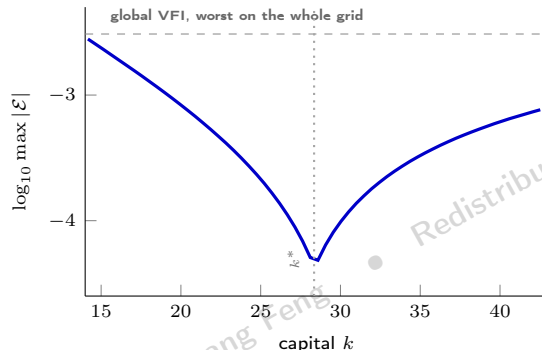
That lag is $\kappa = 0.03475$ doing its work, and it is the classic RBC propagation mechanism: the shock is transitory, the capital stock's response is not.

None of this needed a simulation. From four numbers, the entire dynamic response — and, with two more lines, every second moment in closed form.

And every one of these numbers is **certainty equivalent**. Redo it with σ ten times larger and the picture is identical.

Ledger pert_first_order: peak $\hat{c}$ 0.3697%, peak $\hat{k}$ 0.4811% at quarter 22, shock half-life 13.5 quarters.

Step 4: Grade It, Across the Whole State Space



The Euler residual of the first-order solution, at every capital level in 4.B5's grid.

It is a **V**, centred on k^* , and it is exactly the shape the theory predicts: best where we expanded, worse in both directions, and monotone in the distance.

at k^*	$10^{-4.32}$
at $k = 0.5 k^*$	$10^{-2.55}$
accuracy lost	1.8 decades

Ledger fig_pert_error and three_methods. "One decade" is a factor of ten in the error (3.A1).

The Uncomfortable Comparison

Now put the two solutions side by side on the same model, the same grid, the same residual definition.

	4.B5 grid	1st order
numbers stored	2,100	4
solve time	0.94 s	< 0.001 s
worst $\mathcal{E}$, near k^*	$10^{-2.82}$	$10^{-4.01}$
worst $\mathcal{E}$, whole grid	$10^{-2.51}$	$10^{-2.56}$

Perturbation is 1.2 decades **better** near the steady state, and — read it carefully — it is **not worse anywhere**. Its worst residual on the whole grid, $10^{-2.56}$, essentially matches the grid solver's $10^{-2.51}$.

Ledger three_methods. Both rows use the same unit-free residual on the same random draws.

The wrong conclusion

“So perturbation is at least as good as a global method.” No. What this shows is that **the grid solver is not a gold standard**. Its own error is $10^{-2.5}$, dominated by interpolation and the grid — so it cannot certify anything finer than that.

Comparing two approximations tells you they differ. It does not tell you which is right. You need a *third*, better one — and Block B builds it.

What One More Block of Algebra Buys

Solve the twelve second-order conditions of 5.A5 and re-grade:

	1st	2nd
numbers stored	4	12
solve time	$< 0.001 \text{ s}$	0.02 s
worst $\mathcal{E}$ near k^*	$10^{-4.01}$	$10^{-6.14}$
worst $\mathcal{E}$, whole grid	$10^{-2.56}$	$10^{-4.50}$

Roughly two decades, on both bands, for eight more numbers. That is the best return on effort anywhere in Sessions 3 to 5 — and it also buys the risk correction, which first order reports as identically zero.

Ledger three_methods. Second-order coefficients from the 12-equation residual-matching solve of 5.A5, residual 3.6×10^{-9} .

Second-order perturbation now beats the 2,100-number grid solution **everywhere on the grid**, by two decades, using twelve numbers.

If the state is one-dimensional, smooth, and you care about behaviour near the steady state, this is very hard to beat. All three conditions matter.

What We Deliberately Did Not Do

- **No occasionally binding constraint.** Put a borrowing limit in and the policy has a kink; the expansion does not exist at it (5.A1, 5.A5).
- **No large shocks.** $\sigma = 0.007$ per quarter. 5.A5 measured the σ^2 law failing by 60% at twenty times that.
- **No second endogenous state.** Perturbation scales to fifty states far better than a grid does — its real advantage, and this model cannot show it.
- **No global check.** We graded against residuals and against 4.B5, neither of which is the truth.
- **No analytic derivatives.** Dynare uses them; ours are numerical, which cost six orders of magnitude of residual at second order until the step size was fixed.

The honest summary

Perturbation is extraordinarily good at what it does and gives you **no warning at all** when you leave the region where it does it.

Which is the argument for Block B: not that projection is more accurate here — it is — but that it tells you its own domain.

Takeaways

- The same model as 3.A4 and 4.B5, solved with **no grid, no iteration and no maximization** — a steady state, two matrices, and one QZ decomposition.
- The entire first-order solution is **four numbers**, and $k^* = 28.3484$ agrees with 4.B5 to every digit.
- From those four numbers: the full impulse response. Capital peaks at 0.481% in quarter **22**, from a shock that is half gone by quarter 13.5.
- The residual is a **V centred on k^*** : $10^{-4.32}$ there, $10^{-2.55}$ at half the steady-state capital. **1.8 decades** lost, exactly as the theory says.
- Against 4.B5 it is **1.2 decades better near k^* and no worse anywhere** — which says less about perturbation than about the grid solver, whose own error is $10^{-2.5}$.
- **Second order buys two more decades for eight more numbers**, and beats the 2,100-number grid solution everywhere on it. Comparing two approximations cannot tell you which is right — Block B builds the third one.

Bridge: Keep the Accuracy, Lose the Grid

Block A traded **global validity** for speed and dimension, and we measured exactly what the trade cost: nothing near the steady state, 1.8 decades at half of it, and everything at a kink.

Block B makes the opposite trade. Keep the solution **global** — valid on a whole domain, constraints and all — but stop storing one number per grid point. Store a handful of coefficients instead, and define “solved” as “the residual is zero”.

After the break (5.B1)

Projection: the residual function, and what it means to make a function small. Then bases, Chebyshev, finite elements, collocation against Galerkin — and the same model again, solved a fourth way.

Spoiler, from the ledger: 56 numbers, and **four decades** better than anything in this block.